A Parameter-Free Neutrino Sector from One Geometry (D_IV^5)

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Abstract

Bubble Spacetime Theory (BST) derives the Standard Model from a single geometric object, the bounded symmetric domain $D\_IV^5 = SO_0(5,2)/[SO(5)\times SO(2)]$, specified by five integers (N_c=3, n_C=5, C_2=6, g=7, N_max=137; rank = 2) and zero free parameters. This note states the neutrino sector that follows, as a self-contained, falsifiable theory. Its backbone is a single structural fact — D_IV^5 has rank 2 — which forbids the right-handed neutrino, forces Majorana nature, and gives the neutrino Majorana mass matrix rank 2, hence a one-dimensional kernel: the lightest neutrino is massless, $m_1 = 0$, exactly. With the measured mass-squared splittings as the only external input, the sector is then fully determined: normal ordering, Σm_ν = 58 meV, $0\nu\beta\beta$ effective mass 1.5-3.7 meV, atmospheric angle in the upper octant. Every mixing angle is a ratio of the five integers, each within 1σ of the current global fit. The prediction is welded to BST's no-crossing dark-energy result, which commits the sector to the *tight* cosmological mass bound (< 64 meV) with no escape — making Σm_ν = 58 meV a sharp, near-term falsifiable target.

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1. What BST predicts (read this first)

Quantity	BST	Formula (integers only)	Current data	Agreement
Nature	Majorana	ν_R forbidden (rank 2 / W-21)	($0\nu\beta\beta$ TBD)	testable
Lightest mass m_1	0 (exact)	rank-2 Majorana kernel	<= ~30 meV	prediction
Ordering	Normal	m_1=0 lightest	NO favored ~2.5σ	OK
Σm_ν	58.2 meV	m_1=0 + measured Δm^2	< 64 meV (cosmo)	OK (5.8 meV margin)
**	m_ββ	** ($0\nu\beta\beta$)	1.5 - 3.7 meV	NO band, m_1=0
$\sin^2\theta_{12}$	0.300	N_c/(2n_C) = 3/10	0.307 ± 0.012	0.6σ
$\sin^2\theta_{23}$	0.571 (upper octant)	(n_C−1)/(n_C+2) = 4/7	0.561 ± 0.015	0.7σ
$\sin^2\theta_{13}$	0.0222	1/[n_C(2n_C−1)] = 1/45	0.02203 ± 0.0006	0.3σ
Majorana phases	0 (Dirac-like CP)	rephasing / N=3 strata	unmeasured	testable

Nothing here is fit to neutrino data. The angles are integer ratios; the masses use only the two measured Δm^2 as input (BST supplies m_1=0 and the ordering, not the absolute splittings — see Section 9).

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2. The mechanism: rank 2 forbids ν_R and zeroes the lightest neutrino

The entire sector follows from three forced steps.

(i) No right-handed neutrino. On D_IV^5 the weak SU(2)_L couples only to fermion windings on a non-orientable Möbius locus (K3 / Pin(2)-Z_2; the rank-2 quotient Pin(2)/SO(2) = Z/rank = Z/2). Left-handed (orientation-preserving) windings couple; right-handed cannot. ν_R is not a state of the geometry — it is topologically forbidden, not merely heavy. (This is one of BST's Five Absences: no sterile neutrinos.)

(ii) Therefore Majorana. With no ν_R, the Dirac mass term m_D ν̄_L ν_R cannot form. The only available neutrino mass is the Majorana term m_L ν̄_L^c ν_L. Neutrino nature is forced to be Majorana — a prediction, testable by 0νββ (Section 5).

(iii) Therefore $m_1 = 0$, exactly. The geometric Majorana mass matrix is built on the generation structure of D_IV^5, whose rank is 2. A rank-2 symmetric matrix on three generations has a one-dimensional kernel. That kernel is the lightest neutrino: $m_1 = 0$, exact and radiatively stable (a residual Z_3 generation symmetry protects it). There is no fine-tuning — the zero is the rank deficiency of a rank-2 object acting on three states.

This is the neutrino face of a structure that appears elsewhere in BST identically: the same rank-2 deficiency gives the confirmed Dirac operator on D_IV^5 exactly one ground state (its kernel is one-dimensional; see Section 8). One geometry, rank 2, one unpaired zero mode — in the operator it is the vacuum, in the neutrino sector it is the massless species.

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3. Mass spectrum

With $m_1 = 0$ and the measured splittings $\Delta m^2_{21} = 7.5\times10^{-5}$ eV^2 and $\Delta m^2_{31} = 2.46\times10^{-3}$ eV^2 (normal ordering, which m_1=0 forces):

- $m_1 = 0$
- $m_2 = \mathrm{sqrt}(\Delta m^2_{21}) = 8.7$ meV
- $m_3 = \mathrm{sqrt}(\Delta m^2_{31}) = 49.5$ meV
- Σm_ν = $m_1+m_2+m_3$ = 58.2 meV (58.2-58.8 meV over the current Δm^2 range)

This is the *floor* of the normal-ordering sum; BST predicts the neutrino masses sit exactly on it (m_1=0), not above.

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4. Mixing

The three PMNS angles are ratios of the five integers (Section 1 table). The physically sharp qualitative predictions:

- Atmospheric angle in the upper octant: $\sin^2\theta_{23} = 4/7 = 0.571 > 1/2$. BST predicts θ_{23} is not maximal and lies above 45°. JUNO/DUNE octant resolution is a direct test.
- Reactor angle $\sin^2\theta_{13} = 1/45 = 0.0222$ — the tightest match (0.3σ), a clean integer ratio.
- Solar angle $\sin^2\theta_{12} = 3/10$ (0.6σ).
- CP structure Dirac-like: the Majorana phases are predicted zero (BST's CP is a location/existence statement on the N=3 generation strata, not extra phases). Only the single Dirac δ_CP is physical. A measured nonzero Majorana phase would falsify this.

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5. Neutrinoless double beta decay

Majorana nature (Section 2) makes 0νββ allowed, and m_1=0 with normal ordering pins the effective mass to a narrow band:

|m_ββ| = 1.5 - 3.7 meV (band set by θ_{13}, the mass eigenvalues, and the Dirac phase; Majorana phases predicted zero).

This is a firm, sharp target for next-generation tonne-scale detectors (current KamLAND-Zen / LEGEND sensitivities are ~ 10-100 meV). The falsification structure is two-tiered, and we place the clean kill at the inverted-ordering floor — not at the band edge — because BST's own band, taken to a joint 3σ excursion of the oscillation parameters, extends to ~4.2 meV, so a "~4 meV kill" would be nitpickable:

- Prediction: |m_ββ| = 1.5 - 3.7 meV (central; up to ~4.2 meV at joint 3σ).
- Tension zone: 4 - 15 meV — a signal here is above BST's band but does not cleanly falsify it (it could be accommodated by ordinary parameter excursion or an unforeseen small m_1); it is disfavoring, not decisive.
- Clean kill: |m_ββ| >= ~15 meV (the inverted-ordering floor, ~15-19 meV). A 0νββ signal at this level *requires* inverted ordering or a degenerate spectrum, which directly contradicts BST's forced normal ordering with $m_1 = 0$. This kill is excursion-proof — no choice of oscillation parameters brings BST's NO/m_1=0 band up to the IO floor.

The other side of the two-sided prediction: BST also forbids a null — Majorana nature means 0νββ must eventually appear (no exclusion of the entire NO band down to ~1 meV).

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6. The dark-energy lock (why 58 meV is sharp, not soft)

BST separately predicts no phantom crossing in dark energy: w(a) relaxes to −1 *from above* (w > −1, monotone; the DESI phantom-crossing preference is degenerate with a low-z supernova calibration offset). This is not optional decoration — it removes BST's escape from the tight neutrino bound. The cosmological Σm_ν bound depends on the dark energy model: a phantom (w < −1) loosens it (< 0.16 eV), but ΛCDM/quintessence (w >= −1) keeps it tight (< 0.064 eV). Because BST is committed to no-crossing (w > −1), it is committed to the tight bound < 64 meV and cannot use the loose one.

BST's Σm_ν = 58 meV therefore sits just 6 meV under its own applicable ceiling. The two predictions are welded: one cannot be relaxed to save the other.

Kill condition: a cosmological determination of Σm_ν below ~58 meV (under no-crossing dark energy) falsifies BST. The 2.7σ current preference for *positive* neutrino mass is favorable; the next round of DESI + CMB tightening is the test.

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7. Falsifiers (concrete, near-term)

Prediction	Falsified by	Experiment
Σm_ν = 58 meV (tight bound)	cosmological Σm_ν < ~58 meV under w>−1	DESI + CMB (Planck/CMB-S4)
$m_1 = 0$ / normal ordering	inverted ordering confirmed m_ββ	JUNO, DUNE, atmospheric ν = 1.5-3.7 meV (<=4.2 at 3σ)
Majorana nature	0νββ excluded to the NO band	tonne-scale 0νββ
θ_{23} upper octant	θ_{23} measured in lower octant	JUNO, DUNE
Majorana phases = 0	nonzero Majorana phase	(long-term)
No sterile neutrino	any sterile ν detection	short-baseline (SBN, etc.)

Every row is a two-sided, near-term test. The sector is maximally exposed: with zero free parameters, any single confirmed violation kills the theory.

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8. The through-line: one asymmetry, two sectors

The neutrino theory is one instance of a single geometric fact. D_IV^5 has rank 2, and rank 2 acting on three-fold structure always leaves exactly one thing unpaired. In the Dirac operator credentialed on D_IV^5 this week (the Kostant cubic Dirac operator, confirmed self-adjoint with a discrete climbing spectrum), the unpaired object is the single ground state — the vacuum, kernel dimension one. In the neutrino sector, the unpaired object is the single massless species — $m_1 = 0$. These are two faces of the same rank-2 asymmetry. The lightest neutrino is massless for the same reason the vacuum is unique: *the breaking is the content.*

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9. Honest status (what is derived, what is input)

- DERIVED (from the geometry, no data): Majorana nature; $m_1 = 0$ exactly; normal ordering; the three PMNS angles as integer ratios; Majorana phases = 0; no sterile ν; the dark-energy coupling (tight-bound commitment).
- STRUCTURE + DATA (identified predictions): Σm_ν = 58 meV and |m_ββ| = 1.5-3.7 meV use the two *measured* mass-squared splittings as input. BST forces m_1=0 and the ordering; it does not derive the absolute splittings Δm^2_{21}, Δm^2_{31} or the m_3/m_2 ratio (a direct-geometry attempt at the ratio was tested and does not close — that door is closed honestly, not fudged).
- Not claimed: the absolute neutrino mass scale beyond m_1=0; the individual masses independent of the measured splittings.

The theory's strength is precisely its exposure: it removes free parameters from the neutrino sector down to two measured numbers, pins everything else to five integers, and couples the result to an independent cosmological prediction so it cannot wriggle.

Companion empirical data-sheet (Σm_ν vs DESI, 0νββ vs KamLAND-Zen/LEGEND, θ_{23} octant vs JUNO/DUNE, all current): Grace. Registry: T2563. Toy: 5245 (3/3 PASS, current PDG 2024 / NuFIT 5.2).